Chapter 7

Genome-Scale Metabolic Models: Reconstruction and Analysis

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Abstract

Metabolism can be defined as the complete set of chemical reactions that occur in living organisms in order to maintain life. Enzymes are the main players in this process as they are responsible for catalyzing the chemical reactions. The enzyme–reaction relationships can be used for the reconstruction of a network of reactions, which leads to a metabolic model of metabolism. A genome-scale metabolic network of chemical reactions that take place inside a living organism is primarily reconstructed from the information that is present in its genome and the literature and involves steps such as functional annotation of the genome, identification of the associated reactions and determination of their stoichiometry, assignment of localization, determination of the biomass composition, estimation of energy requirements, and definition of model constraints. This information can be integrated into a stoichiometric model of metabolism that can be used for detailed analysis of the metabolic potential of the organism using constraint-based modeling approaches and hence is valuable in understanding its metabolic capabilities.

Key words: Genome-scale metabolic network reconstruction, Metabolic networks, Metabolic flux analysis, Flux balance analysis, Constraint-based modeling

1. Introduction

The vast amount of available genome sequence databases for organisms and the development of high throughput bioinformatics and experimental techniques have led to intense efforts to functionally characterize these genomes and build genome-scale datasets for a variety of organisms like *Neisseria meningitidis* (1). Genome-scale metabolic network models are an example of this. Whereas enzyme kinetics are often unknown, the stoichiometry of the reaction the enzyme catalyzes is in most cases well established. The reaction stoichiometry of the biochemical reactions that are present forms

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the basis of a genome-scale network in which the different metabolic reactions are connected through their common substrates and products. Whether a metabolic reaction can take place inside an organism can primarily be reconstructed using the information provided in the annotated genome combined with information from curated databases and literature. Such a genome-scale network thus represents the metabolic potential of the organism and hence is valuable in understanding its metabolic capabilities. Genome-scale metabolic models have been built for several organisms (2, 3) and have been used to analyze cultivation data, get a better understanding of cellular metabolism (1, 4), develop metabolic engineering strategies (5–8), design media and processes (1, 7, 9, 10), assess theoretical capabilities (2), and even for online control of the process (11), which illustrates its usefulness for process development and optimization. In this chapter, an introduction to genome-scale metabolic network reconstruction and the constraint-based modeling approach is presented.

2. Materials

For the genome-scale metabolic network reconstruction and constraint-based modeling methods described below, the following software materials can be used.

1. Major online bioinformatics databases (see Table 1).
2. Software for metabolic network analysis (see Table 2).
3. Tools for assigning putative gene function, e.g., BLAST or InParanoid.
4. Automated tools for metabolic network reconstruction, e.g., Pathologic (Pathway Tools) and AUTOGRAPH.

3. Methods

3.1. Metabolic Network Reconstruction

Before a metabolic model for a given microorganism (or specific phenotype) can be used to gain new insights into its metabolic capabilities or evolutionary history, it must first be built from the available (scattered) genomic, biochemical, and physiological information. This process is known as model reconstruction and has been extensively reviewed before (12, 13). The process of model reconstruction is schematically shown in Fig. 1.

It starts with the functional annotation of a sequenced genome, meaning that for each open reading frame (ORF) a function is sought using literature and pathway databases. Part of the functionally annotated genes will code for enzymes involved in (primary) metabolism.

Table 1
Major online databases that can be used as resource for model reconstruction

Database	URL	Description
GOLD	http://www.genomesonline.org	Monitoring of genome sequencing projects around the world
TIGR	http://www.tigr.org	Information about genomes of sequence organisms
NCBI	http://www.ncbi.nlm.nih.gov/Genbank	Contains sequence data of microbial and higher organisms
EBI	http://www.ebi.ac.uk/embl	Similar as GenBank
KEGG	http://www.genome.ad.jp/kegg	Database that includes all microorganisms with publically available genome sequence. Holds both genomic and metabolic information
BioCyc	http://biocyc.org	Similar as KEGG, also includes MetaCyc
Expasy	www.expasy.org	Protein sequence database that provides organism-specific annotation
EMP	http://www.empproject.com	Generic information on enzymes and metabolism
BRENDA	http://www.brenda.uni-koeln.de	Information about enzymes (organism related)
UniProt	http://www.uniprot.org	Universal Protein Resource database
MIPS	http://mips.gsf.de	Information about genomic structure and integration of data for several organisms
TCDB	http://tcdb.ucsd.edu	Classification system for membrane transport proteins
TransportDB	http://www.membranetransport.org	Predictions of membrane transport proteins for fully sequenced genomes
Reactome	http://www.reactome.org	Curated database of biological pathways
BiGG	http://bigg.ucsd.edu	Repository of reconstructed genome-scale metabolic models
TargetP	http://www.cbs.dtu.dk/services/TargetP/	Predicts subcellular location of eukaryotic proteins
SignalP	http://www.cbs.dtu.dk/services/SignalP/	Predicts the presence and location of signal peptide cleavage sites in amino acid sequences from different organisms
PSort	http://psort.hgc.jp	Similar as TargetP

Table 2
Software for metabolic network analysis

Package	URL	Reference
Flux balance analysis/metabolic flux analysis		
CellNetAnalyzer	http://www.mpi-magdeburg.mpg.de/projects/cna/cna.html	(71, 72)
MetaFluxNet	http://mbel.kaist.ac.kr/lab/mfn/index.html	(73, 74)
FBA, Cobra Toolbox	http://gcrg.ucsd.edu/Downloads/Flux_Balance_Analysis http://gcrg.ucsd.edu/Downloads/Cobra_Toolbox	(75)
CellDesigner	http://celldesigner.org/index.html	(76, 77)
Fluxor	http://sourceforge.net/projects/fluxor/	
Simpheny (commercial)	http://www.gtlifesciences.com/technology/Simpheny.html	(78)
Isotope-based metabolic flux analysis		
FiatFlux	http://www.imsb.ethz.ch/researchgroup/nzamboni/Software/fiatflux	(79)
13C flux	http://www.13cflux.net/omix/	(80)
OpenFLUX	http://web.aibn.uq.edu.au/cssb/Resources.html	(81)
Elementary flux modes/extreme pathways		
Metatool	http://pinguin.biologie.uni-jena.de/bioinformatik/networks/	(82)
Yana	http://yana.bioapps.biozentrum.uni-wuerzburg.de	(83, 84)
ExpA (extreme pathways)	http://gcrg.ucsd.edu/Downloads/Extreme_Pathway_Analysis	(85)

For these enzymes the reaction stoichiometry is obtained as well as information on the reversibility of the reaction, which can next be included in the model. In case of different cellular compartments also the localization of the enzyme can be determined. A special reaction is the reaction for biomass formation, which is a lumped reaction, not connected to a single enzyme but the sum of many biochemical reactions. This results in a basic reaction set, which usually still contains errors like a wrong reaction stoichiometry and pathway gaps, which must be manually corrected. Next, the model must be experimentally validated, which includes estimation of the unknown energy parameters, being the efficiency of ATP generation and the amount of ATP required for maintenance and biomass formation. This process is iterative meaning that in case the model does not agree with experimental data, the model is critically evaluated again starting at the functional annotation. The different aspects of the model reconstruction are discussed in more detail below. The process of reconstruction of the (genome-scale) metabolic network of a microorganism requires a significant input

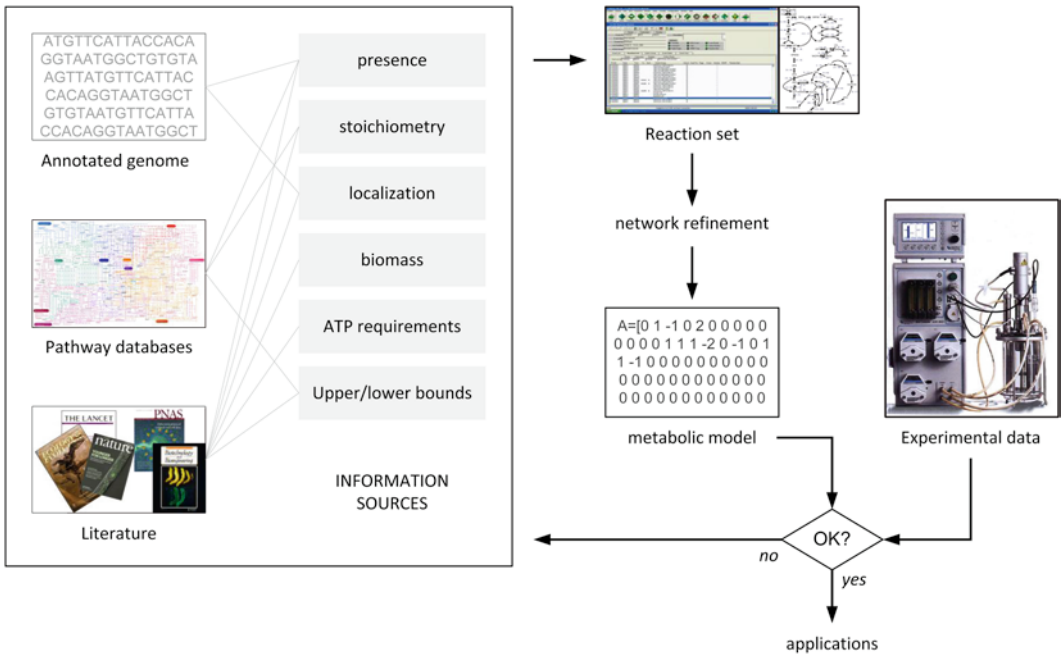

Fig. 1. The iterative process of metabolic network reconstruction. The process starts with gathering the current knowledge about the microorganism from multiple sources of information. Next the construction and debugging of the reaction set, the building of a stoichiometric matrix, and the comparison of the *in silico* simulation results using experimental data are carried out. Once the experimental data (i.e., measured rates) are in agreement with the *in silico* predictions, the model can be used for further applications like, e.g., development of metabolic engineering strategies. Adapted from Rocha et al. (69).

from online bioinformatics databases, where information regarding genome sequencing and metabolic reactions can be found. Some important online resources are listed in Table 1.

3.2. Identification of Reactions and Their Stoichiometry

An important requisite in genome-scale model reconstruction is the availability of a well-annotated genome. Meaning that the process of genome assembly, identification of ORFs, using a good gene prediction training set and their localization on the genome (structural annotation), and the assignment of possible biological roles (functional annotation) should have been done carefully. In all cases, detailed inspection of the genome (functional) annotation is required prior to or concurrently with reconstruction of the network. For this, reliable databases, such as GOLD, TIGR, NCBI and EBI (Table 1) can be consulted. Important data to be extracted from the various databases include gene or ORF, assigned function(s), (translated) sequence and for protein coding genes the Enzyme Commission (EC) number that corresponds with the reaction(s) encoded by the gene. In the case where an ORF has no function assigned, the sequence can be analyzed aiming at assigning a putative function.

Using orthology-based methods like BLAST or Inparanoid (14), putative functions can be found and based on the similarity score between orthologous genes one can decide to include the associated reaction in the model, hereby assigning the putative function. However, caution is required when including putative functions, especially if shortened genes are identified, since some gene regions coding for specific protein domains that are related with the annotated function can be lost. In some cases, genes encoding proteins with multiple functions may lose the domain(s) responsible for some of these activities. These circumstances might lead to the assignment of a gene to an incorrect functional category, which could lead to an erroneous determination of the set of functions an organism is able to perform. Next to the inclusion of genes or ORFs encoding enzymes (with an EC number), also membrane transport reactions (that lack an EC number) should be included. Hence, annotated genomes should also be analyzed for transporter functions. Genes involved in regulatory processes of metabolic reactions or signal transduction cascades are currently excluded from model reconstruction. Nevertheless, it is useful to store information related to these processes as they might be interesting for future exploration.

The result of the above mining of the genome is a set of genes with an identified function. Notably, in *Neisseria meningitidis* strain MC58 the percentage of protein coding genes with a known function varies from 54 to 60% (the latter value includes putative functions), meaning that many of the genes still have an unknown function. Metabolic network reconstruction is time-consuming and automated methods to accelerate the process have been proposed (15–18). Two examples are Pathologic, which is part of Pathway Tools (17) and AUTOGRAPH (15), both of which can be used to predict as much metabolic information as possible for a given species. The Pathologic algorithm takes annotated genomes as input and predicts gene–reaction associations based on name-matching and EC-codes with the idea to curate the predictions afterwards (17). The EC-code matching approach to link metabolic information to genes is similar to other methods like metaSHARK, IdentiCS, and Pandora (19–21). Notably, Pathologic is the first step in the construction of a so-called PGDB (pathway-genome database) which consists of gene–reaction associations.

Several organism-specific PGDBs have been constructed and manually curated (9, 22–26). Despite the significant time saving using the automated methods, manual curation and inspection is still required to ensure elimination of possible errors and inconsistencies. Once the genome has been adequately mined, databases that describe gene-reactions (e.g., KEGG, BioCyc) and relevant biochemical literature should be consulted for determination of the exact reaction stoichiometry including cofactor requirements, reaction reversibility/irreversibility and consistency by inspection

of the elemental balances of each reaction. When it is unclear whether NADH or NADPH is used as cofactor in a particular reaction, most times both reactions are included. In this case, the name of the gene stays the same but the names of the reactions differ. Notably, in some cases the cofactor specificity can be determined from the sequence and phylogenetic analysis (27). Similarly, when an enzyme accepts more than one substrate, the gene is associated with several reactions. For reactions that are catalyzed by enzyme complexes, several genes are associated with one (or more) reactions. In such cases, the overall or net reaction is often included. Isoenzymes are encoded by different genes but catalyze the same reaction(s) and should be considered separately. In order to determine whether a reaction is reversible or irreversible, the standard Gibb's free energy change associated with the reaction can be used (28). This reversibility/irreversibility information leads to definition of appropriate thermodynamic model constraints on the direction of reactions. Finally, like for the overall modeling approach the reconstruction process (Fig. 1) can also be seen as an iterative process where information obtained from several sources is combined to construct a preliminary set of reactions and constraints that are then analyzed to detect potential errors, for example, erroneous reactions in terms of stoichiometry, errors related to duplication of metabolite names or gaps in biochemical pathways.

3.3. Compartments and Localization

Another important aspect for metabolic network reconstruction is the definition of compartments and the assignment of the localization of enzymes inside these compartments. Software tools like SignalP, TargetP, and PSort (Table 1) deduce such information from the amino acid sequence of the enzymes. If a particular metabolite can be present in different compartments (e.g., by translocation or biochemical reactions), it is important to distinguish between these identical metabolites by using compartment-specific naming (e.g., M_mit for metabolite M in the mitochondrion and M_cyt for the same metabolite M in the cytosol). The translocation of such metabolites between compartments should be included as separate reactions in the model.

3.4. Biomass Formation

The next step in genome-scale model reconstruction is to add the reactions that describe the formation of biomass. The biomass formation reaction can be described by the assembly of macromolecules that constitute the biomass, e.g., protein, RNA, DNA, lipids and carbohydrates, which in turn are formed from precursor metabolites, e.g., amino acids, nucleotides, fatty acids, and sugars. It is also possible to define a reaction where biomass is directly formed from the precursor metabolites. The reaction stoichiometry is determined by the composition of the biomass and the biopolymers. For example, the reaction equation for protein can be directly derived from the measured amino acid composition. This illustrates

the importance of accurate determination of the biomass composition. The biomass composition of several well-studied organisms like *Escherichia coli* (29, 30), *Neisseria meningitidis* (1, 4), *Lactococcus lactis* (31), *Saccharomyces cerevisiae* (32–38), *Penicillium chrysogenum* (39), and *Pichia pastoris* (40) is available. An important component in the biomass synthesis reaction is the number of ATP molecules that need to be included per unit of biomass, i.e., ATP stoichiometry, as described below.

3.5. Network Refinement

Once the complete set of reactions has been extracted and the stoichiometry has been checked the network may still contain gaps due to annotation errors, incomplete or incorrect EC-codes or annotation errors in protein databases. In addition, some reactions proceed spontaneously without being catalyzed by an enzyme. It is important to correct these errors by manual investigation of the network using biochemical literature, comparative genomics approaches (41), transcriptomics data (42) or by direct measurements. In addition, automated approaches like Pathologic can be used.

3.6. Energy Requirements

Although the ATP stoichiometry coefficients of many reactions that include ATP are known, with respect to ATP generation in mitochondria or the bacterial membrane and ATP usage for growth and maintenance the stoichiometry is not fixed and must be estimated from experiments. Part of the energy required for growth is accounted for in the model, like for example the ATP required to polymerize amino acids into protein. However, for another part related to, for example, the assembly of the biopolymers into functional biomass, and growth-related turnover of macromolecules and maintenance of gradients over the membrane this is not known. This part of the required energy, also called the growth-associated maintenance requirements represents all energy consumption related to growth not accounted for in the model, which must be determined experimentally. The same holds for the nongrowth-associated maintenance requirements for ATP, which represents the ATP required to keep the organism alive. ATP consumption for nongrowth-associated maintenance is usually included in the model by a separate irreversible reaction that converts ATP into ADP and orthophosphate. Growth and nongrowth-associated maintenance values of *Neisseria meningitidis* (4) and other organisms can be found in the literature (30, 43) or determined experimentally (4, 44).

In addition to the growth- and nongrowth-associated energy requirements also the ATP stoichiometry of oxidative phosphorylation is not fixed and is unknown. As mentioned earlier, the plasma membrane of bacteria contains ATP synthases that catalyze the formation of ATP in response to a transmembrane proton gradient. This gradient is formed when electrons are transferred from mostly NADH or $FADH_2$ to oxygen, where in the case of NADH more

protons are translocated than for $FADH_2$. The amount of ATP formed per oxygen atom (P/O ratio) thus depends on the ratio of NADH and $FADH_2$, and the translocation stoichiometry (H^+/ATP). Thus to obtain the right flux of ATP formation in oxidative phosphorylation determination of the P/O ratio (45, 46) is important. The unknown parameters being the P/O ratio, and the growth- and nongrowth-associated parameters can be estimated from experiments with different carbon substrates or ratios of mixed substrates for pseudo steady-state growth at different growth rates as described extensively elsewhere (44).

3.7. Metabolic Network Analysis

When the pathway gaps and other possible errors have been fixed and the stoichiometry of all reactions is defined, the set of reactions can be used as a metabolic model to describe the function of the metabolic network in a quantitative manner. There are three types of metabolic models:

1. Steady-state models, which only take into account the flow of metabolites through the system in (pseudo) steady state.
2. Steady-state kinetic models, which take into account the flow of metabolites through the system in (pseudo) steady state and contain at least one kinetic equation that relates the concentration of a metabolite to a reaction rate.
3. Dynamic kinetic models, which take into account the kinetics of all different enzymes involved in the reaction network that are valid under dynamic conditions.

In an ideal situation, the model contains different levels of information, including reaction stoichiometry, reaction kinetics and regulatory information. Despite several studies in this direction (47–52), the required kinetic and regulatory information is currently lacking. For this reason a (pseudo) steady-state approximation, using the reaction stoichiometry in combination with mass balancing of the fluxes (1, 4, 30, 45, 53–57) is generally applied. The metabolic capabilities of the constructed network may be calculated using constraint-based computer simulation methods like metabolic flux analysis (MFA), flux balance analysis (FBA), elementary flux modes (58), or extreme pathways (59). An overview of available flux analysis software packages is provided in Table 2 and a brief summary of modeling techniques (i.e., MFA and FBA) is provided below (see Note 1).

From the set of reactions, including the transport reactions over the membrane, a stoichiometric matrix, S ($m \times n$), is constructed where m is the number of metabolites and n the number of reactions. Each column of S specifies the stoichiometry of the metabolite in a given reaction from the metabolic network. The construction of a network is relatively easy for small networks. However, if the networks become more complex mistakes are easily made. The consistency of a network may be checked by looking at

the elemental balances for each reaction. The elemental composition is written in an elemental composition matrix E, with columns containing the different compounds and rows containing the different elements. Since each reaction must obey the law of elemental conservation, the following equation should always hold:

$$E \cdot S = 0 \tag{1}$$

Next, the mass balance for each intracellular metabolite can be constructed resulting in:

$$\frac{\mathrm{d}C}{\mathrm{d}t} = S \cdot v \tag{2}$$

where C is the vector of concentrations of intracellular metabolites (mole.(unit biomass)$^{-1}$), v is the vector of conversion rates (mole. unit biomass^{-1}.h^{-1}), including the transport rates over the cell membrane. Since metabolism operates on a much faster time-scale than for instance cell division, it is reasonable to assume that metabolic dynamics have reached a so-called pseudo steady state, where metabolite concentrations are constant in time or, in other words, that there is no accumulation of the intracellular metabolites (law of conservation of mass). Hence, their net accumulation rate is zero giving:

$$S \cdot v = 0. \tag{3}$$

From the above (3) the solution space, i.e., all possible combinations of flux values that satisfy the equation, can be derived. If a flux does not contribute to this solution space meaning its value is always zero, the reaction is called a dead end. Dead ends can never carry a flux and are usually caused by a gap in a pathway or a wrong annotation. Usually part of the transport rates over the membrane, being the specific consumption and production rates of substrates and products are estimated from measurements resulting in the next equation.

$$S_c \cdot v_c = S_m \cdot v_m \tag{4}$$

where v_c contains the unknown and v_m contains the measured fluxes. S_c and S_m are the corresponding parts of the matrix S. If (4) has one unique solution, the system is called determined. The unknown fluxes can then be calculated from the measured rates using the next equation in case S_c is a square matrix:

$$v_c = -S_c^{-1} \cdot S_m \cdot v_m \tag{5}$$

where S_c^{-1} is the inverse of the matrix. In case the matrix S_c is not a square matrix, the inverse cannot be calculated and the following equation can be used:

$$v_c = -(S_c^T \cdot S_c)^{-1} \cdot S_c^T \cdot S_m \cdot v_m \tag{6}$$

where S_c^T is the transpose of matrix S_c. Calculation of fluxes based on measurements is called MFA. However, often the system of equations is not determined and a unique solution of (6) does not exist. This can be due to two reasons. First of all insufficient rates are measured (see Note 1). Second, even if all rates that can be measured are measured it is still possible that a unique solution does not exist due to the presence of different pathways with the same overall stoichiometry, i.e., parallel and cyclic pathways (see Note 2). For both cases S_c^{-1} and $(S_c^T \cdot S_c)^{-1}$ cannot be calculated. For both situations usually part of the fluxes can be calculated from the measurements, while in principle at least two of the fluxes cannot be calculated from the measurements. In case of insufficient measurements, the problem can of course be simply solved by measuring the additional fluxes. For the second situation this is not possible. In some cases ^{13}C MFA, which uses ^{13}C labeling distribution of intracellular metabolites, can provide additional constraints and can yield a determined model system for this situation for which a unique solution can be calculated (60). Whereas this is the preferred method to estimate the unknown fluxes, it is also experimentally and mathematically complex and time-consuming. Another option is using the so-called Moore–Penrose pseudo inverse the solution given by:

$$v_c = -S_c^{\#} \cdot S_m \cdot v_m \tag{7}$$

where $S_c^{\#}$ is the (left) pseudo inverse of matrix S_c. For the determined part of the network, this equation calculates the flux values. For the under-determined parts that do not have a unique solution, this equation calculates the minimum norm solution. The minimum norm solution is only one of the infinite possible solutions and minimizes the sum of the fluxes squared (see Note 1). However, the biological relevance of this solution remains questionable. To get a biologically more meaningful solution, linear optimization can be used for both mentioned situations (see Notes 1 and 2). Within the solution space defined by the connectivity (stoichiometry) and capacity constraints (upper and lower boundary values of fluxes and transport rates lb and ub, respectively), a solution is calculated based on a defined objective function. This approach is called FBA. For this, constraints are defined for the different flux values, including the transport rates over the cell membrane and an objective function is defined resulting in the next definition of the problem (see Note 1):

$$\begin{aligned} &\text{Objective function}: && \text{maximize}(c \cdot v_i) \\ &\text{Constraints}: && S \cdot v = 0 \\ & && \text{lb}_i \le v_i \le \text{ub}_i, i = 1, \ldots N \end{aligned} \tag{8}$$

where c.v_i contains the combination of fluxes to be optimized (usually a single flux).

The main objection against the technique of linear optimization (i.e., linear programming) to find a solution for an under-determined network is that the objective function may not be valid for the biological system and one should realize that a mathematical solution is calculated that may not represent the actual situation. Using FBA, the original solution space of the under-determined parts of the model system is reduced to a single solution. Notably, this single solution is not necessarily unique and there can be more than one flux distribution that reaches the optimal value of the objective function.

3.8. Objective Functions and Constraints

Some examples of objective functions described in the literature are maximization of biomass formation, maximization of ATP production, minimization of glucose uptake, and minimization of the production of NADH and NADPH (56, 61, 62). Successful prediction of growth rate and metabolite secretion rate has been done using the FBA approach combined with maximization of biomass formation as an objective function (63–66). Nevertheless, as stated, the calculated intracellular flux distribution in these cases may not represent the true values. An extensive comparison of FBA-based *in silico* flux predictions, using different objective/constraint combinations, with in vivo fluxes from ^{13}C-experiments demonstrated that prediction of flux distributions is, within limits, possible. The essential element in this case is to identify constraints and the most relevant objective for a specific condition, since no single objective predicted the experimental data for wild-type *E. coli* under all conditions (62). Constraints can be set on certain enzymatic reactions on the basis of information found in the literature or determined experimentally. For example, measured production and consumption rates can be included as constraints on the transport fluxes over the cell membrane. The main constraints to be added to the set of reactions are the so-called thermodynamic constraints, which are related to the reversibility/irreversibility of the reactions. Such thermodynamic information can be found in online databases (e.g., Brenda, UniProt, Kegg, and BioCyc) but should be used with caution since the information in the databases is not organism-specific. It is worth noting that a software tool like anNET (28) can be used to test quantitative data sets for thermodynamic consistency.

4. Notes

1. Flux analysis in a nutshell
   Consider the simplified metabolic network in Fig. 2 in which a substrate (S) can be converted into biomass (B) and a by-product (P) through several chemical conversions with a stoichiometry of one, and which involves five intracellular metabolites (Mi).

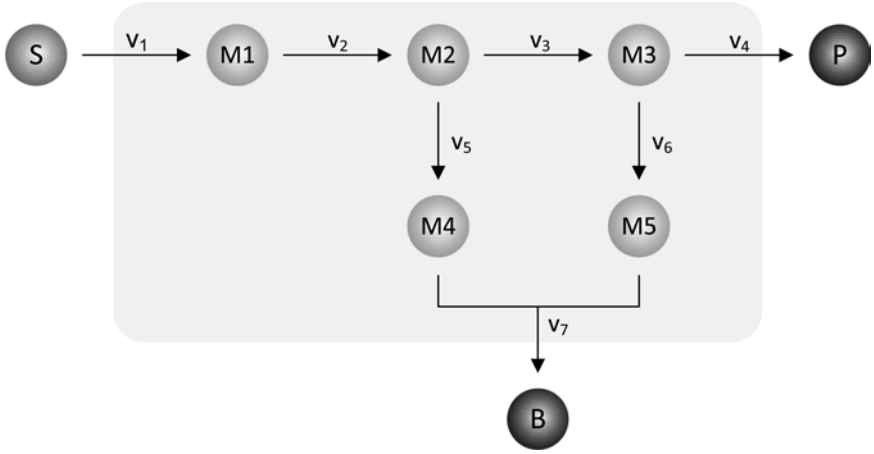

Fig. 2. Simplified metabolic network. Substrate (S) can be converted into biomass (B) and product (P) through several chemical conversions involving intracellular (*light grey* area) metabolites (Mi).

In the network in Fig. 2, there are five intracellular metabolites and seven reactions (five equations and seven unknowns). Therefore, the nullspace (i.e., solution space) is two-dimensional (Fig. 3a). If values for the biomass and product formation rate (v_7, v_4) are known from measurement, the degree of freedom of the network is zero, which means that the system is determined and a single unique solution for all fluxes and the exchange rates can be calculated (Fig. 3b) using the linear-algebra techniques described in the text. A special situation occurs when v_1 is also measured. In this case, more rates are measured than is needed for solving the system. In this case, the measurements are called redundant and a procedure called data reconciliation can be used to check the consistency of the measurements. This is described in detail elsewhere (67, 68). Notably, solving determined and redundant metabolic networks is often referred to as MFA. In the case when no measurements are available, the system is under-determined. A solution can be calculated (i.e., estimated) by assuming that the system fulfills an optimality condition. By using a particular objective function, for example maximizing the product formation rate (v_4), the optimal solution (i.e., optimal flux distribution) that optimizes this object function can be found using linear programming. To find a particular maximum, one or more constraints are required, otherwise the solution to the optimization will be unbounded (Fig. 3c). The technique of solving an under-determined metabolic model is commonly referred to as FBA and is often used in genome-scale models, in which the number of unknown fluxes is too large to calculate from measurements (57). Measurements of uptake and production rates are then added as constraints on the transport fluxes over the membrane.

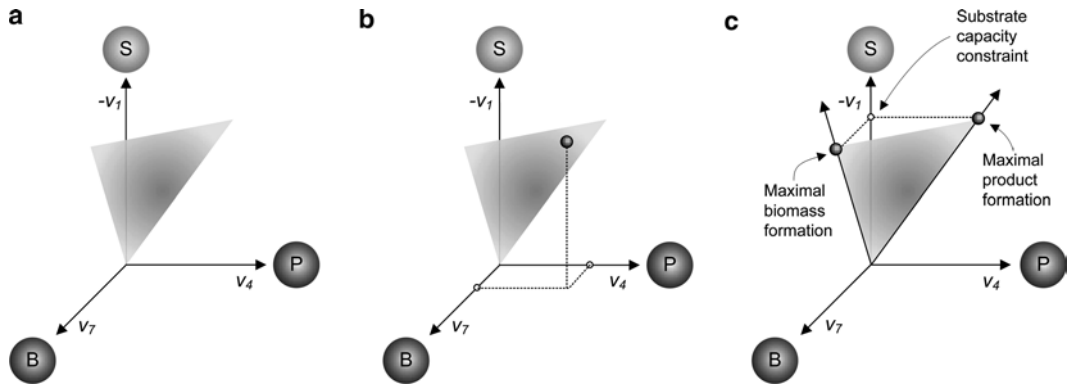

Fig. 3. Metabolic flux and flux balance analysis (FBA). (**a**) The null space of the example network is two-dimensional. (**b**) Metabolic flux analysis (MFA): if consumption and production rates are measured a unique solution can be calculated. (**c**) FBA: if only the substrate conversion rate (S) is known from measurement, a solution that optimizes a particular objective, like maximize biomass formation or maximize product formation can be calculated using linear programming. Adapted from Teusink et al. (57).

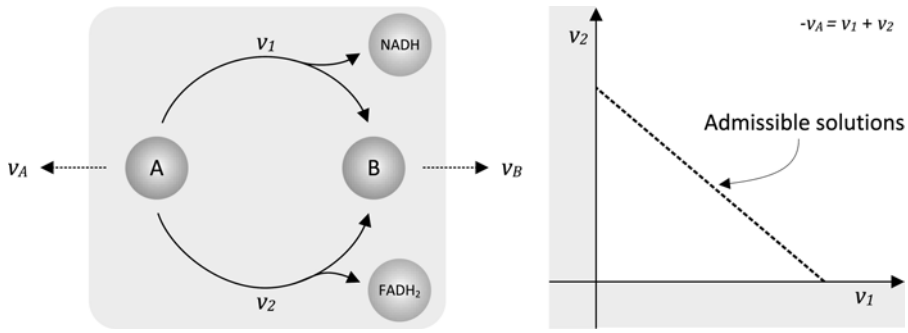

Fig. 4. Example of an under-determined network. Adapted from Bonarius et al. (70).

2. Inherently under-determined networks

In the simple under-determined metabolic network shown in Fig. 4, compound A is consumed by a cell and is metabolized to compound B via two different irreversible routes and then secreted. One route produces redox potential in the form of NADH, the other in the form of $FADH_2$.

The only flux balance in this system is that the sum of the two internal fluxes must equal the exchange flux. Once v_A is known from measurement, the solution space forms a straight line in the v_1 v_2 plane. Even if all rates are measured, a unique flux solution does not exist. Since the fluxes are constrained to be positive, we can only be in the positive quadrant of the plane, and thus the solution space is the dashed line shown in Fig. 4. A solution obtained using the Moore–Penrose pseudo-inverse,

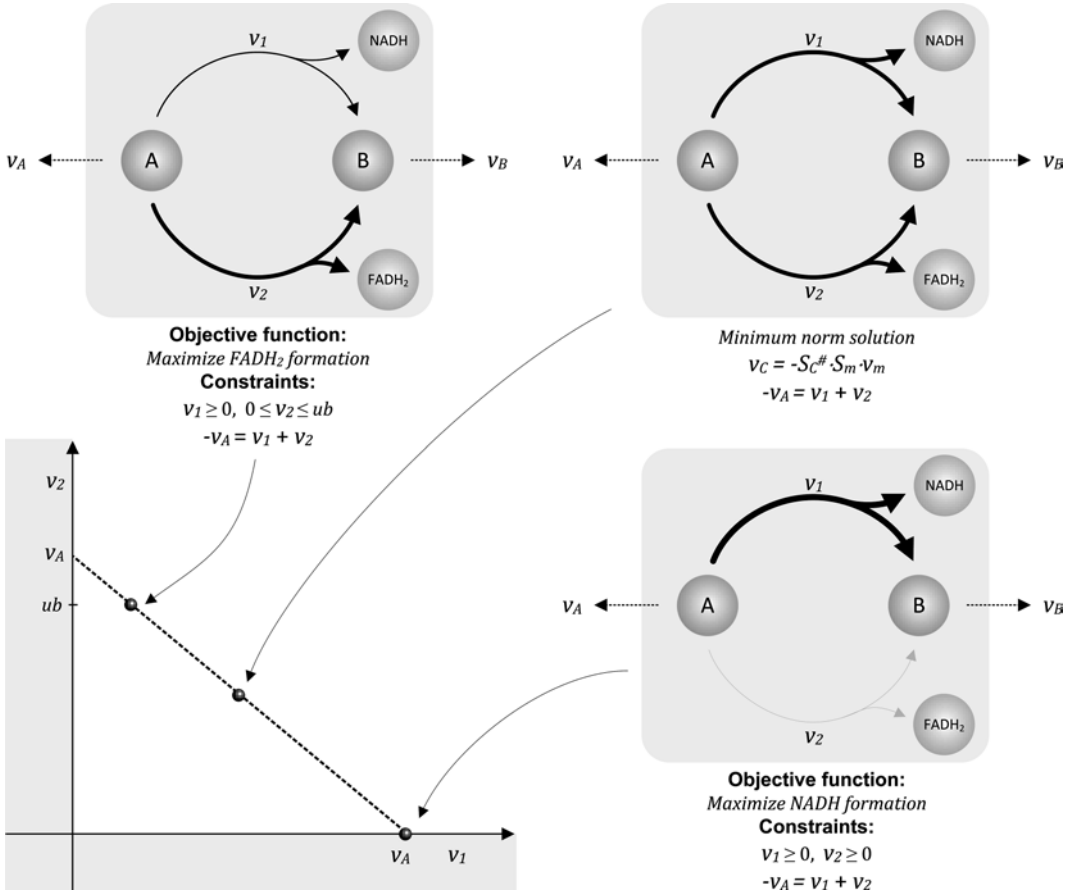

Fig. 5. Solutions of the example network. The minimum norm solution (*upper right*) adapted from Bonarius et al. (70), the solution obtained using linear programming and the objective function maximize NADH formation and constraints $v_1 \geq 0$ and $v_2 \geq 0$ (*lower right*) and the solution obtained using linear programming and the objective function maximize $FADH_2$ formation and constraints $v_1 \geq 0$ and $0 \leq v_2 \leq ub$ (*upper left*).

also called the minimum norm solution (or minimum norm constraint), minimizes the length of the solution vector without any further restrictions, i.e., it minimizes the sum of the squares of the fluxes. The Moore–Penrose pseudo-inverse gives the one solution vector that has the smallest Euclidian norm (e.g., the smallest square of its length) and is shown in Fig. 5. In linear programming, one uses an objective function (and constraints) to calculate an optimal result. If one uses the objective function that maximizes NADH production, it is clear that v_2 should go to zero and v_1 to the maximum value equal to the uptake rate. That solution lies at the extreme point of the solution space to the right (Fig. 5).

Conversely, if one maximizes the redox production from this metabolite in the form of $FADH_2$, and the upper boundary

value for v_2 equals ub (e.g., obtained from measured maximal enzyme activity), then the optimal solution is at the opposite end of the solution space (Fig. 5). These simple examples illustrate how optimal solutions in linear spaces are found at the extremities of the allowable solution space.

To determine which elements of v_c have a single solution and to determine the relationship between the infinity of solutions for the other elements of v_c, the nullspace of S_c can be calculated. The nullspace or solution space or kernel matrix is defined as the set of linear independent vectors v_n that fulfill the equation:

$$S_c \cdot v_n = 0. \tag{9}$$

The number of independent nullspace vectors is equal to the number of columns in S_c minus the rank of S_c. Each nullspace vector can be added an arbitrary number of times to the base solution given in the above equation. Thus the complete solution is:

$$v_c = -S_c^{\#} \cdot S_m \cdot v_m + \text{nullspace}(S_c) \cdot \lambda \tag{10}$$

where λ is a vector with as many elements as there are vectors (columns) in the nullspace of S_c. For all possible values of λ, the solution remains valid. In other words, the nullspace vectors contain combinations of reactions of which the sum is always zero.

Parallel pathways have a common net reaction and cause under-determinancies in the metabolic network. To mathematically detect parallel pathways, they are considered to run in the opposite direction so that the net production rate is zero. These are called nullcycles and the best way to identify these nullcycles is to determine the nullspace of S. The number of nullspace vectors (i.e., columns) equals the number of nullcycles. If a unique solution is wanted, one of the parallel pathways must be removed or at least one reaction rate should be measured (e.g., by using ^{13}C labeled substrate) or depending on the conditions, fluxes should be set to zero. Identification of nullcycles is often called a network sensitivity check and the number of parallel pathways in a metabolic network is a measure of its robustness. It is worth noting that futile cycles are not the same as parallel pathways as they have ATP consumption as net effect.

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